

BrainyMinis: Interactive Learning Web Application based on Emotional Interaction Design Framework

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ABSTRACT

The "BrainyMinis" web application serves as an interactive, web-based pedagogical tool for young children, concentrating on fundamental mathematical skills. At present, traditional teaching methods such as worksheets and chalk-and-talk are utilized, which frequently lead to diminished engagement among children. A survey with preschool teachers (n=90) revealed that the current teaching method is lacking in interactivity, engaging resources, and progress monitoring tools, which are necessary to measure each child's growth effectively. Thus, BrainyMinis is proposed as an interactive web-based learning application that incorporates flashcards, assessments, and puzzle-based activities to support the acquisition of abstract mathematical concepts, such as number recognition and basic operations, through structured play-based learning. The platform systematically records learners' mathematics performance and progress toward predefined achievement targets, enabling progress monitoring by parents or guardians. As a result, learning outcomes become more observable and measurable through embedded assessment mechanisms. To evaluate usability, engagement, and overall system acceptance, User Acceptance Testing (UAT) was conducted with preschool educators and parents. The results indicated a high level of user satisfaction in terms of interface usability, content relevance, and perceived learning effectiveness. BrainyMinis seeks to address gaps in early mathematics instruction by providing a culturally responsive and adaptable learning environment that supports cognitive development and fosters sustained engagement in foundational learning.

KEYWORDS: Interactive learning, Preschool children, Foundational mathematics, Progress tracking, Safe-paced learning

Received 5 November 2025; Revised 29 January 2026; Accepted 6 March 2026

Doi: <https://doi.org/10.59953/paperasia.v42i1b.1070>

1. INTRODUCTION

Early exposure to mathematics is critical for cognitive development and future academic achievement. Techniques such as chalk-talk, teacher-directed instructions, and printed materials, including worksheets, are ineffective in guiding young children and have little to no impact on developing an appreciation and understanding of mathematical concepts (Ompok, 2021; Siew & Tee, 2024). Observation reveals a strong tendency to use worksheets and most of the teacher's instructions, suggesting that active student engagement does not positively affect learning (Siew & Tee, 2024). All these traditional methods employed in preschools, characterized by passive learning and memorization, fail to address young children's learning dispositions as they grow and learn, and consequently lead to low mathematical skill development among Malaysian preschool children.

The traditional chalk-talk approach in Malaysia often fails to engage young learners or adapt to their learning paces, limiting their ability to grasp mathematical concepts effectively (Ghazali et al., 2023). Although multimedia-based learning has been found to increase students' engagement (Nowawi & Ahmad, 2023), there are still gaps in game-based, interactive mathematics resources for preschoolers in Malaysia. For example, Lê et al. (2024) found that young children solved only about two-thirds (66.36%) of simple single-digit addition problems correctly overall, with some items showing accuracy near 50–58%, signalling that foundational arithmetic skills are still developing in the early years. Preschool teachers in Malaysia face significant challenges in effectively teaching mathematics, including a shortage of interactive resources, difficulties in managing diverse learning paces, and a lack of assessment tools to track individual progress (Siew & Tee, 2024). Children aged 4 to 6 thrive in play-based and

self-paced environments, where they can learn through engaging, interactive activities (Babakr et al., 2019). However, the existing teaching aids in Mathematics learning do not encompass the type of interaction that would enable children to engage in their curiosity and learn without necessarily considering it a grind (Sabri et al., 2023; Zulkefli & Hashim, 2025).

Interactive learning is essential for early childhood education, especially in core subjects like mathematics, which are vital for cognitive development. Studies show that children with early exposure to mathematics tend to perform well academically in later years, not only in math and reading but also in writing classes (Sabri et al., 2023). Additionally, introducing abstract mathematical concepts enhances critical thinking and problem-solving skills throughout a child's education (Rusdawati, 2019; Sabri et al., 2023). Although many teachers acknowledge the value of play-based and student-centered learning (Muryadi & Mohd Hashim, 2025), a persistent gap exists between their beliefs and actual classroom practices, with most still relying on traditional, teacher-directed methods that limit student engagement and creativity (Wood, 2014; Siew & Tee, 2024). This disconnect is often due to systemic barriers such as curriculum demands, standardized assessments, and insufficient professional training (Edwards, 2017; Pyle & Danniels, 2017). As a result, students miss out on the cognitive, social, and emotional benefits associated with active, exploratory learning (Bodrova & Leong, 2015; Hassan & Obaydullah, 2020). To address this challenge, educators and institutions must adopt innovative, evidence-based strategies such as enhanced technology and gamified environments that support more dynamic and participatory learning experiences (Verenikina et al., 2016).

The new and innovative learning platforms have demonstrated effectiveness in addressing similar issues. For instance, Nowawi and Ahmad (2023) used flashcards, quizzes, and storybook press interventions via web-based applications in teaching Malay to preschoolers. Their study revealed that children value interaction designs and multimedia elements, which can reduce distractions and consequently enhance children's interest and understanding. Similarly, BrainyMinis has incorporated the effective use of interactive multimedia to promote numeracy at the literacy level for young children. BrainyMinis is also based on previous studies, such as the interactive multimedia courseware on Mental Arithmetic Techniques (Ahmad et al., 2015), which aims to enhance problem-solving skills and accuracy. Since multimedia generally combines text, graphics, sound, and interactive elements, it is especially effective in maintaining children's attention. Stakeholders suggest that learners are more engaged with activities related to daily life and learning through games, as multimedia enhances interest and reduces

dropout rates (Nowawi & Ahmad, 2023).

BrainyMinis is a web-based application developed to support early childhood mathematics education by addressing constraints associated with conventional instructional approaches. The system is designed as a supplementary learning platform that integrates interactive digital activities with culturally contextualized content relevant to the Malaysian educational setting. A nature-themed interface featuring Malaysian wildlife, including the Malayan tiger and orangutan, is incorporated as part of the learning environment to provide contextual support alongside mathematical instruction. The application consists of several instructional components, including interactive games, flashcards, quizzes, and puzzle-based activities. These components are structured to introduce and reinforce foundational mathematical concepts while allowing learners to engage with content at a self-directed pace. The system records learner interactions and performance data, which are used to support real-time progress tracking and the identification of learning milestones.

BrainyMinis includes a monitoring module that enables parents and teachers to access learner progress information. This module is intended to support observation of individual learning patterns and facilitate timely instructional support when required. For educators, the system provides basic performance evaluation tools derived from recorded activity data, allowing for the review of individual student progress over time.

The primary objectives of the BrainyMinis system are to (1) provide an interactive digital environment for early mathematics learning, (2) support progress monitoring by parents and educators, and (3) facilitate performance evaluation in early childhood education contexts. By combining interactive instructional elements, contextualized thematic design, and systematic progress tracking, BrainyMinis is positioned as a digital learning system intended to complement early mathematics instruction within the Malaysian context.

2. RELATED WORK

Several existing solutions were examined to better understand how adaptive learning and gamification help researchers identify effective design strategies and enhance early childhood learning platforms. For instance, AdaptedMind has gained significant attention for its interactive and personalized teaching approach.

AdaptedMind is a web-based educational platform that incorporates game-based learning and adaptive instructional strategies to support student learning. The system allows learners to progress through content

along individualized learning paths, with lesson difficulty and focus areas adjusted based on learner performance. Gamification elements, such as point systems and achievement indicators, are integrated to support sustained learner engagement. The platform provides interactive instructional materials, including quizzes, videos, and story-based activities, covering subjects such as mathematics, reading, social-emotional learning (SEL), social studies, and science. In mathematics, the instructional design emphasizes step-by-step problem-solving processes, while literacy components focus on vocabulary development and reading comprehension through narrative-based content. AdaptedMind aligns with standards-based learning objectives but does not adhere to a specific national curriculum. Additionally, the platform includes progress monitoring features that allow parents and teachers to review learner performance and provide instructional support as needed.

This web application provides an adaptive learning paths customize lessons according to student performance metrics, enabling differentiated support. User engagement is facilitated through built-in gamification mechanisms, including point systems, achievement badges, and reward structures. The platform also incorporates interactive instructional materials, such as video-based explanations and formative assessments, which promote active knowledge acquisition and concept reinforcement. Furthermore, the system supports progress monitoring through parent and instructor access, allowing stakeholders to evaluate learning outcomes and identify areas requiring intervention. Despite these features, the application presents certain limitations. It is not explicitly aligned with national or region-specific curricular standards, potentially reducing its applicability in education systems with rigid compliance requirements. Additionally, the absence of collaborative learning functionalities, such as peer interaction, discussion forums, or group-based activities, limits opportunities for social learning and cooperative skill development.

Komodo Math is a UK-developed digital learning platform designed to support learners in Key Stages 1 and 2 by strengthening foundational mathematics skills. The platform aligns with the UK national curriculum and seeks to address learning gaps through structured, short-duration practice sessions. Instruction is delivered through 20-minute lessons conducted three to five times per week, following a "little and often" pedagogical approach intended to support skill retention and progressive mastery. The program begins with a diagnostic assessment to identify individual learning needs and inform personalized learning pathways. Its instructional design draws on principles from cognitive psychology, such as spaced practice and retrieval, to support long-term retention of mathematical concepts.

Learning content is organized into sequenced, individualized lessons that emphasize mathematics fluency and mental calculation skills. Engagement is supported through a reward-based system, including "Komodo Belts," which functions as a progress indicator rather than a motivational device. The platform also provides parent-facing dashboards that enable monitoring of learner progress and performance over time. Personalized learning plans are overseen by UK-qualified teachers to ensure appropriate pacing and content alignment with learner ability. Despite these strengths, Komodo Math has notable limitations. Its curriculum framework is not aligned with the Malaysian KPSK syllabus, which may restrict its adoption in educational contexts that require strict adherence to national standards. Additionally, while the platform supports individual practice and progress tracking, it offers limited opportunities for collaborative or interactive learning, such as peer engagement or group-based activities, which could enhance learner participation and broader educational outcomes.

2.1 Malay Language Learning for Kindergarten Students through an Interactive Web-based Application

Nowawi and Ahmad (2023) also focus on developing a multimedia-based web application for teaching the Malay language to preschoolers. The study employs multimedia learning theory, which integrates text, images, sound, and video to enhance cognitive learning and engagement. This approach is designed to improve language acquisition, social skills, and problem-solving abilities in young learners by providing interactive and engaging content. The application's development follows the Web Development Life Cycle (WDLC), consisting of five stages: planning, data collection and analysis, design, development, and testing. During the planning phase, learning goals are defined, and the needs of the target audience are assessed. In the design phase, user-friendly interfaces are created, while development utilizes PHP, JavaScript, HTML, CSS, and MySQL for backend functionality and student progress tracking. The testing phase involves evaluating the application's functionality with real students and teachers, revealing areas for improvement, such as adding more quiz content and incorporating audio into the flashcards. Overall, this study emphasizes the use of multimedia learning, the WDLC methodology, and interactive design as key components for creating an effective educational tool that enhances language learning in early childhood.

One of the key strengths of the system is its use of multimedia to enhance cognitive engagement, combining text, images, audio, and video to improve language retention among preschoolers. The interactive nature of the content fosters a positive and engaging learning environment, supporting the development of early language skills. Additionally, the self-paced

learning feature enables students to progress at their own pace, catering to diverse learning styles and providing a more personalized experience. Another strength is the real-time progress tracking system, which enables parents and educators to monitor each child's development, allowing for timely interventions to address learning gaps effectively. Despite these strengths, the system has some limitations. One key weakness is the limited variety of learning activities. While multimedia is utilized effectively, the educational approach could benefit from a broader range of interactive elements, such as more dynamic visual and auditory learning tools. Although multimedia content is available, the audio-visual integration could be improved. For example, adding more interactive audio elements, such as pronunciation guides or contextual dialogues, would strengthen language learning. Furthermore, the user interface (UI) could be more intuitive and engaging for young learners. Some past systems have struggled with complex navigation or visual elements that are not fully optimized for preschoolers, so enhancing the UI to be more user-friendly and visually appealing is a vital area for improvement.

2.2 Enhanced Interactive Mathematical Learning Courseware Using Mental Arithmetic for Preschool Children (iMLc 2.0)

Ahmad et al. (2015) describe iMLc2.0, an interactive multimedia application designed to teach preschoolers addition and subtraction using mental arithmetic techniques. Addressing the limitations of traditional methods, iMLc2.0 uses text, images, audio, video, and animation to create an engaging learning environment that incorporates finger-brain and mental-imagery approaches to calculation. A three-phase study (analysis, design/construction, and evaluation) rigorously tested the application's effectiveness. Experimental tests with preschoolers showed significant improvements in calculation speed and accuracy after using iMLc2.0. User acceptance testing with teachers yielded highly positive results, indicating strong perceived usefulness and ease of use. These findings strongly support the application's potential as a valuable educational tool. The successful implementation of iMLc2.0 suggests the potential for similar multimedia-based approaches in teaching mathematics to other age groups, with appropriate modifications to align with different developmental stages.

However, this study does not detail specific computational algorithms; instead, it focuses on the pedagogical design and evaluation of the system. The core pedagogical method is mental arithmetic, employing techniques like the finger-brain and mental-imagery approaches, which are learning strategies rather than formalized algorithms. The courseware's design relies heavily on multimedia learning principles, aiming to reduce cognitive load and enhance learning

by presenting information through multiple channels (visual and auditory). This aligns with theories such as Mayer's multimedia learning, but is not presented as a formal algorithm. The interactive elements, such as drag-and-drop activities and quizzes, are guided by constructivist learning theories emphasizing active participation. The evaluation methodology employs a classic pre-test/post-test design to assess the impact of iMLc2.0. This involves comparing pre- and post-intervention scores using statistical analysis (like t-tests) to determine significant improvements. Additionally, a user acceptance test uses questionnaires to gather data on Perceived Ease of Use, Perceived Usefulness, Attitude, and Intention to Use, analysing the results using descriptive statistics. These evaluation methods adhere to established research practices rather than employing unique algorithms.

Meanwhile, Piaget's theory of cognitive development informs the design choices, ensuring the product's suitability for the preschool age group. Cognitive load theory, particularly as discussed by Mayer, guides the multimedia design to optimize germane cognitive load while minimizing extraneous cognitive load. Therefore, the paper focuses on applying established pedagogical and evaluation frameworks rather than developing novel algorithms. The "algorithms" present are more accurately the structured processes of designing, implementing, and evaluating an educational intervention. The "iMLc2.0" system for preschool learning offers several strengths, including its use of multimedia (videos, animations, and text), which engages learners with diverse learning styles and enhances effectiveness. It also incorporates mental arithmetic techniques, such as the "finger-brain" method, which fosters cognitive development and critical thinking in young children. The system is user-friendly and has demonstrated positive educational outcomes. However, some weaknesses could be improved. The system's focus on basic arithmetic (addition and subtraction) may limit its scope, as it neglects more advanced mathematical concepts. Its reliance on technology may also create accessibility issues in areas with limited resources. Furthermore, the methods used might not suit all learners, particularly those who need a more structured or diverse approach. The system could benefit from personalized learning features to better address individual needs and improve outcomes for all students.

3. PRELIMINARY STUDY

3.1 Study Design and Methodology

A preliminary needs assessment was conducted to identify user requirements and limitations of existing early mathematics teaching tools, with the aim of informing the design of the BrainyMinis web application. The study employed a cross-sectional quantitative survey design, which is appropriate for capturing educators'

perceptions and practices at a single point in time.

The participants comprised 90 preschool teachers from various regions across Malaysia, recruited using convenience sampling through professional educator networks and online communication platforms. Participation was voluntary, and respondents were required to have current teaching experience with preschool-aged children.

Data were collected using a structured online questionnaire consisting of closed-ended items measured on Likert-type scales. The survey instrument was designed to capture four main aspects:

- a. challenges in current mathematics teaching practices,
- b. engagement levels of existing instructional methods,
- c. availability and usability of interactive learning tools, and
- d. perceived importance of progress monitoring and assessment features.

The questionnaire items were adapted from previous studies on early childhood educational technology and teaching effectiveness (Rasalingam et al., 2014) to ensure content relevance and clarity. The collected data were analysed using descriptive statistical methods, including frequencies and percentages, to summarise teachers' responses and identify dominant trends related to instructional challenges and system requirements.

3.2 Findings of the Preliminary Study

The survey results revealed several key challenges in current preschool mathematics instruction. A total of 52.22% of respondents reported difficulties in accommodating children with differing learning paces, indicating a lack of support for self-paced learning within existing teaching approaches. In terms of engagement, nearly 70% of teachers indicated that interactive games were effective in sustaining children's attention, while 58.89% perceived traditional methods, such as worksheets and chalk-and-talk instruction, as insufficiently engaging.

Progress monitoring emerged as a significant concern, with 67% of respondents emphasising the importance of having tools that enable both teachers and parents to track individual learning performance and development. In addition, 53.33% of teachers reported limited access to interactive educational tools, and 60% expressed dissatisfaction with the usability and engagement level of currently available solutions. Collectively, these findings indicate gaps in interactivity, adaptability, usability, and learning analytics within existing early mathematics instructional tools used in preschool settings.

3.3 Implications for System Design

The outcomes of the preliminary study provide empirical justification for the development of BrainyMinis as an interactive and user-centred learning platform. The identified challenges highlight the need for features that support self-paced learning, engagement through interactive activities, and systematic progress tracking accessible to both educators and parents. By grounding system requirements in actual classroom needs, the preliminary study ensures that BrainyMinis is aligned with pedagogical realities and user expectations. These findings serve as a foundation for subsequent design and development phases, ensuring that the proposed system addresses practical limitations observed in existing early childhood mathematics instruction.

4. SYSTEM ARCHITECTURE

The architecture is composed of four main layers: the user interface (UI), the communication layer, the web server layer, and the data layer.

Figure 1 illustrates the system architecture of BrainyMinis, an interactive web-based learning application designed to support early childhood education. Each component plays a critical role in ensuring seamless interaction, secure data management, and responsive content delivery across different user types. The frontend is developed using HTML5, CSS3, and JavaScript, enabling an intuitive, browser-based interface that is responsive to the unique roles of users, namely Children, Parents, Teachers, and Administrators. Each user type is provided with a tailored dashboard offering role-specific functionalities. All data communication between the frontend and backend is secured via HTTPS, ensuring the confidentiality and integrity of user data, including personal information and learning progress. The communication between frontend activities and backend services is facilitated by a RESTful API, which serves as the bridge connecting GDevelop5-based educational activities with PHP-driven server logic and the database. The API handles essential functions such as user authentication, data submission (e.g., activity scores, progress tracking), and content delivery. These interactions enable the system to record and monitor performance in real-time, allowing teachers and parents to track individual learning trajectories. For the gamification component, GDevelop5 is integrated into BrainyMinis to design interactive educational modules such as puzzles, flashcards, and quizzes, which are developmentally appropriate for preschool learners. This integration supports the system's goal of enhancing engagement, interactivity, and motivation with game-based learning strategies. Overall, the modular and layered architecture of BrainyMinis ensures scalability, maintainability, and a robust foundation for delivering an engaging and adaptive learning experience tailored to the developmental needs of young children.

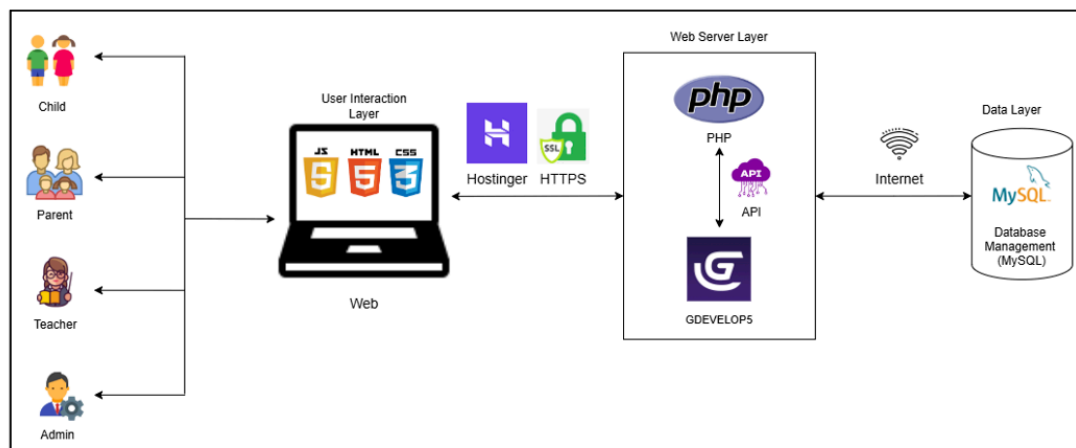


Figure 1: Architecture diagram of BrainyMinis

Figure 1 also presents the overall system architecture of BrainyMinis. The UI layer is implemented using HTML5, CSS3, and JavaScript and operates within standard web browsers. The interface supports multiple user roles, such as Children, Parents, Teachers, and Administrators, each of which is assigned a role-specific dashboard. These dashboards provide access to functions such as activity selection, progress viewing, user management, and content administration, depending on the assigned role.

The Communication layer manages data exchange between the frontend and backend components. All client-server communication is conducted over HTTPS to ensure secure transmission of user credentials, activity records, and system data. A RESTful Application Programming Interface (API) facilitates interactions between frontend processes and backend services. The API supports core operations including user authentication, retrieval of learning content, submission of activity results, and synchronisation of progress records.

The Web Server layer is implemented using PHP and is responsible for handling application logic and processing requests received through the API. This layer manages session control, validation of user inputs, and coordination between learning activities and the database. Interactive learning modules developed using GDevelop5 are integrated at this layer, enabling the execution of game-based components such as puzzles, flashcards, and quizzes. These modules communicate with the server via API calls to store performance data and retrieve configuration parameters.

The Data layer consists of a structured database used to store user profiles, activity metadata, assessment results, and progress records. The database supports persistent data storage and retrieval, enabling longitudinal tracking of user activity and system usage.

Data access is restricted to authorised processes to maintain data integrity and consistency. Overall, the layered architecture supports separation of concerns between presentation, communication, application logic, and data management. This structure facilitates system maintenance, future extension, and integration of additional modules.

5. THE PROPOSED SYSTEM: BRAINYMINIS WEB-APP

BrainyMinis was designed to support young children's learning through an interface that emphasises usability, engagement, and basic safety considerations. The UI applies principles from the Emotional Interaction Design Framework (Chen, 2022) by integrating affect-supportive elements such as child-friendly colour palettes, character-based cues, and playful feedback alongside clear navigation structures. In particular, visual emphasis, readable labels, and simplified screen layouts were used to reduce cognitive load and to help users (both parents and children) interpret functions with minimal instruction. The login/sign-in and authentication pages are primarily intended for parents or guardians. Accordingly, the login and registration forms adopt a conventional structure with clearly labelled fields and straightforward input requirements to facilitate quick access. At the same time, the inclusion of child-oriented visual motifs signals the system's intended target audience and maintains consistency with the overall aesthetic of the learning environment. This combination aims to balance adult usability (efficient access and clarity) with child-appropriate visual identity.

For children users, the main dashboard provides direct access to learning modules via prominent action buttons, enabling children to initiate activities with minimal navigation steps. Progress indicators (e.g., a progress bar and summary chart) are included to provide performance visibility and immediate

feedback, which aligns with the framework's emphasis on reinforcing motivation through timely responses. In addition, short character-based messages are used as affective prompts to acknowledge progress and encourage continuation, supporting sustained attention during repeated learning sessions. The activity screens were implemented using a consistent layout to support learnability across multiple task types. However, interaction components vary according to the learning activity. Quizzes employ multiple-choice response buttons with enlarged selectable areas to accommodate young children's motor control. Flashcards retain similar visual organisation but incorporate simple text-entry components where applicable. Puzzle activities utilise drag-and-drop interactions, requiring children to manipulate elements into correct positions. Across these activity types, the interface maintains large interactive controls, simple navigation, and immediate system feedback, while

character-based visuals function as reinforcement cues following correct actions. **Figures 2 to 11** present the key interface screens and illustrate how the Emotional Interaction Design Framework was operationalised within BrainyMinis

Figure 2 illustrates the main interface of this web application, serving as the main entry point. It offers users straightforward access to register a new account or log in to an existing one. This facilitates parents and guardians in efficiently initiating the system for their children's educational activities.

Once logged into the child's account, the child's dashboard (**Figure 3**) displays all available learning modules and shows their learning progress through daily and overall progress bars. It also highlights any rewards the child has earned, keeping them informed about their achievements.

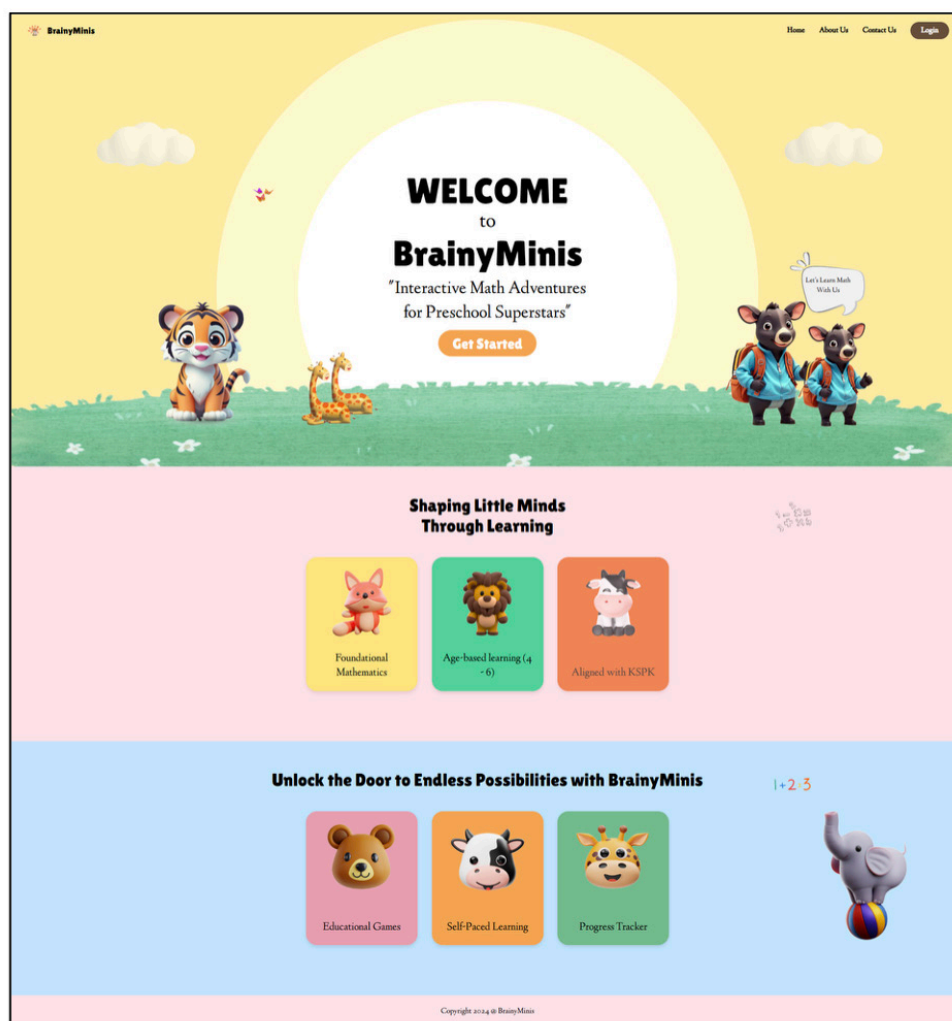


Figure 2: Homepage screen in the BrainyMinis application

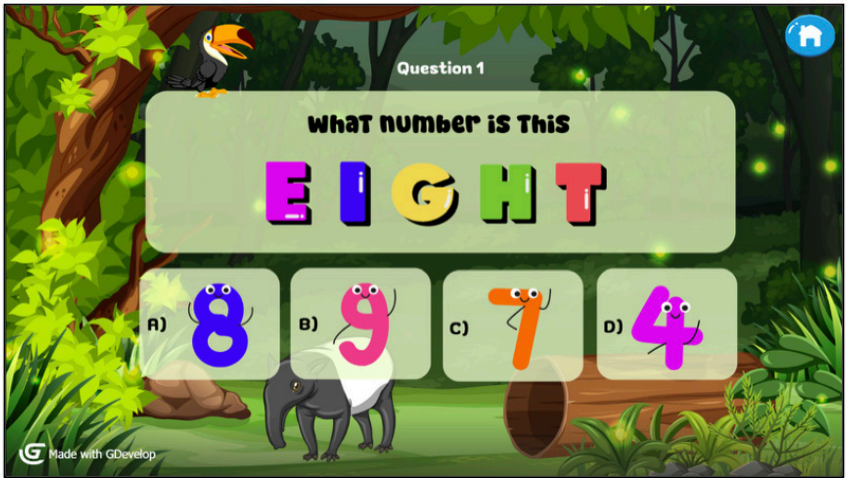


Figure 5: Activity quiz screen in the BrainyMinis application

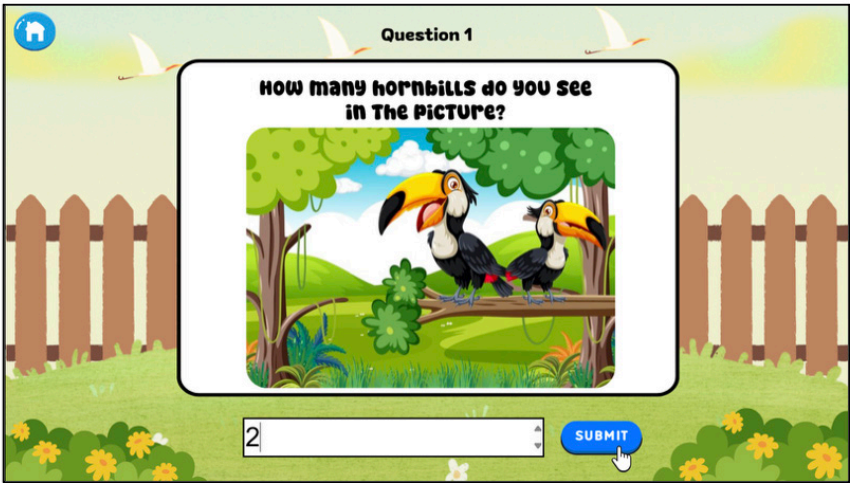


Figure 6: Activity flashcard screen in the BrainyMinis application

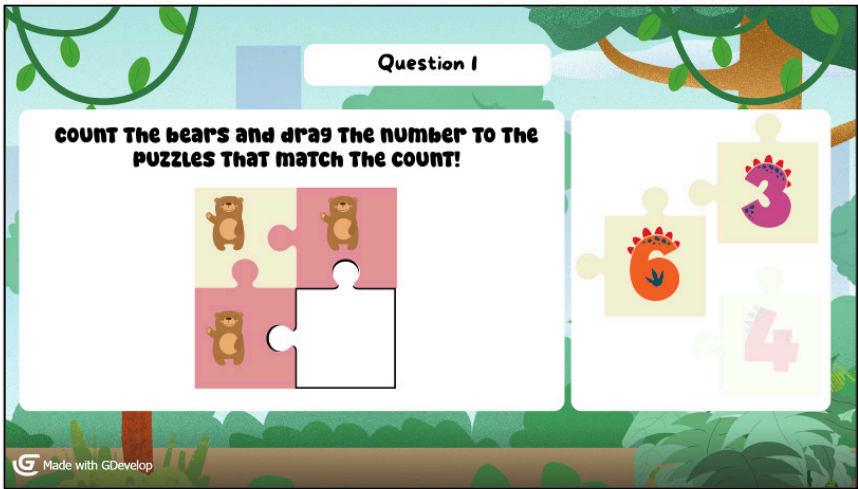


Figure 7: Feedback screen for correct answers in the BrainyMinis application



Figure 8: Feedback screen for incorrect answers in the BrainyMinis application

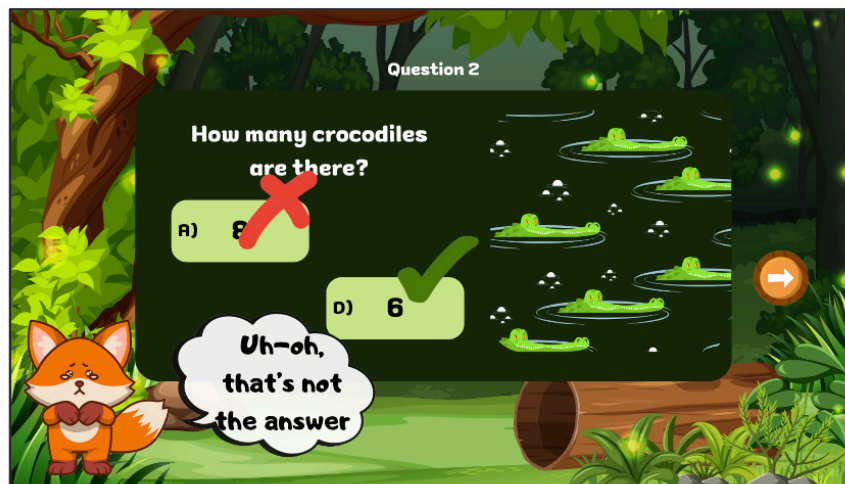


Figure 9: Feedback screen for incorrect answers in the BrainyMinis application

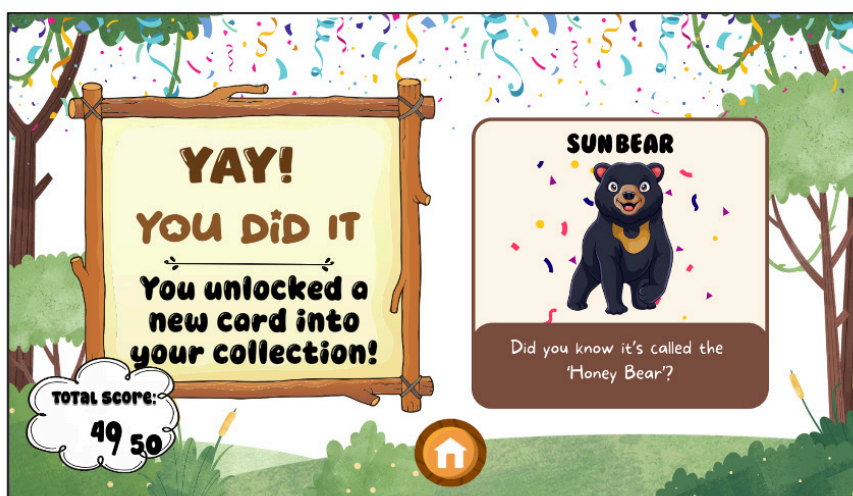


Figure 10: End-of-activity screen in the BrainyMinis application

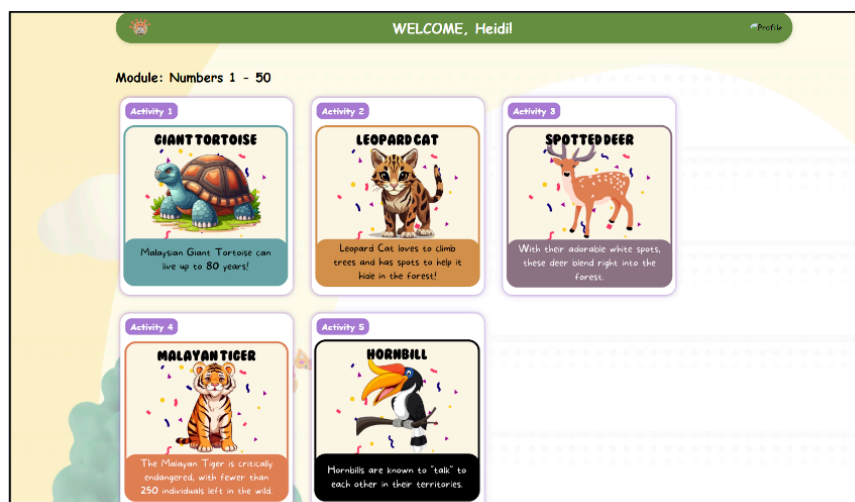


Figure 11: Reward screen in the BrainyMinis application

The BrainyMinis web application utilizes the Emotional Interaction Design Framework to revolutionize early mathematics education for toddlers by directly tackling the negative feelings linked to conventional teaching methods, such as boredom and disengagement. It fosters a positive emotional environment through interactive features like flashcards and puzzles, transforming learning into a delightful and captivating experience, which subsequently nurtures curiosity and interest. Additionally, the application's capability to monitor performance and goal attainment instills a sense of achievement and pride in children while offering parents reassuring insights into their child's development. This encompassing strategy guarantees that BrainyMinis not only imparts crucial math skills but also lays the groundwork for a lifelong passion for learning, effectively closing the gap in early education with a fun, culturally aware, and flexible design. **Figure 12** shows the Emotional Interaction Design Framework for BrainyMinis.

6. BRAINYMINIS SYSTEM EVALUATION

To ensure BrainyMinis functions reliably and meets user expectations, a thorough multi-level testing strategy has been implemented. This approach encompasses unit testing, integration testing, system testing, and user acceptance testing, each designed to evaluate specific aspects of this web application's performance and usability. The first level, unit testing, focuses on the smallest, most basic functional parts of the system. Each module, including learning activities and user management features, is tested independently to identify and resolve issues before integration. For example, the quiz activity will be tested to verify that it accurately processes user choices, provides immediate feedback, and records progress. Likewise, the registration and login features will be validated to

ensure the proper handling of both valid and invalid inputs, such as incorrect usernames or passwords, with appropriate system responses. Next, integration testing checks if different system components work together smoothly. This includes verifying how educational modules, such as quizzes, flashcards, and puzzles, interact with the progress tracking system. It makes sure user actions within these modules are correctly recorded and reflected in the child's learning profile. Integration testing also verifies the communication between the frontend interface and the backend database, ensuring that data, such as scores and timestamps, flows properly across components. After this, system testing evaluates this web application. This stage confirms that BrainyMinis meets all specified functional and non-functional requirements. End-to-end scenarios are run to simulate real user experiences from creating accounts and navigating the dashboard to completing activities and receiving feedback. The goal is to assess whether the application functions as intended by supporting learning activities, recording user interactions, and providing accessible performance data for parents and teachers. Through this structured testing process, BrainyMinis aims to deliver a reliable, user-focused educational solution that aligns with technical standards and the developmental needs of its users.

6.1 User Acceptance Testing

In addition to system-based testing, BrainyMinis was evaluated through User Acceptance Testing (UAT) to assess its functional performance, usability, and appropriateness for use by young learners within an authentic classroom context. This UAT took place in May 2025 at a kindergarten in the Klang Valley region. The primary goal of this phase was to conduct a formative usability evaluation, identifying and addressing interaction and usability issues at an early stage, rather than aiming for statistical generalization across all



Figure 12: Emotional interaction design framework for the BrainyMinis web application

kindergartens. In formative usability testing, it is common to use small samples because the number of newly discovered usability problems tends to be found (Lewis, 1994; Nielsen, 1994; Faulkner, 2003).

Given the age and cognitive development of the target users, UAT did not include quantitative usability measures, such as Likert-scale questionnaires or System Usability Scale (SUS) ratings. BrainyMinis is designed for preschool children, most of whom are unable to provide reliable self-reports, and therefore, structured surveys cannot be completed accurately. The most recent research focused on the interactions between children and computers suggests that the usability evaluation of software aimed at young children should neglect numerical ratings as behavioural indicators of usability, the level of task engagement, and emotional engagement (all of which are extremely relevant in any early childhood scenario) offer a much more accurate picture of usability in the early years (Lehnert et al., 2022). Hence, this UAT can be considered as formative usability testing and as such is based on direct observation of interaction behaviour and feedback from teachers as proxies of adults.

6.1.1 Participant Recruitment

For participant recruitment, a purposive convenience sampling method was used to select participants from a kindergarten that matched the intended deployment context of BrainyMinis (children learning through guided digital activities in a classroom). The specific site was chosen because it represented the real learning environment for the potential users. Subsequently, the kindergarten practiced close supervision and maintained consistent testing conditions, and it also provided feasible access for working safely and ethically with young respondents, an important requirement in child participants (Markopoulos & Bekker, 2003; Barendregt et al., 2006).

Only six kindergarten students ($n=6$) participated in the UAT because the goal was early-stage usability problem detection, where research shows that small samples are usually sufficient to identify a significant portion of usability issues, with diminishing returns after a small number of users (Lewis, 1994; Nielsen, 1994). Additionally, the effectiveness of small-sample testing tends to be stronger when users are fairly similar; if user subgroups differ greatly, more participants are needed (Caulton, 2001). In this specific study, the UAT deliberately focused on a fairly homogeneous group (children from the same classroom environment), which supports the suitability of $n = 6$ for initial acceptance and refinement.

Recruitment was organized through the kindergarten administration and the class teacher, who acted as the gatekeeper. Children were selected based on the target-user criteria (enrolled kindergarten learners capable of participating in brief supervised sessions and following basic instructions). Ethical approval was obtained from the kindergarten head, and parental or guardian consent was secured before the session. During testing, children were supervised by the teacher and could stop at any time, following recommended practices for usability testing with children to ensure comfort and minimize fatigue or stress (Markopoulos & Bekker, 2003; Barendregt et al., 2006). Only one kindergarten was used for the UAT to reflect the real classroom environment, maintaining the same setup and procedure with consistent teacher support, classroom flow, and device arrangement. Past studies emphasize that evaluations with children require careful control of context and facilitation to produce reliable observations and ensure ethical safeguarding. Early-stage studies often start in a single controlled setting before expanding to multiple sites. (Markopoulos & Bekker, 2003; Barendregt et al., 2006). Therefore, the findings from this UAT are best interpreted as initial acceptance evidence and a refinement step, with

future evaluation recommended across additional kindergartens to test broader variability (e.g., different teaching styles, demographics, and device availability) (Caulton, 2001; Faulkner, 2003). Before the UAT, the researcher demonstrated to the children how to use Brainyminis. They were shown how to navigate the modules, answer questions, and use the mouse. After the demo, the children began interacting with the web application independently. The teacher was available to help by translating questions for the children. During the UAT, the researcher closely observed the children as they used the application, noting their engagement, navigation skills, and understanding of the activities. After the UAT was completed, the researcher collected feedback from the teacher regarding its usability, engagement, and alignment with the syllabus. UAT provided valuable insights into the usability and effectiveness of BrainyMinis.

6.1.2 UAT Findings

BrainyMinis was designed to support learning through emotionally engaging interaction, and the UAT analysis focused not only on task completion but also on observable affective cues during use (e.g., smiling/laughter, excitement vocalizations, sustained attention, "again" requests, hesitation, frustration, and help-seeking). This approach aligns with the child–computer interaction study showing that children's technology experiences should be evaluated based on both usability and affective experience ("fun", involvement, emotions) rather than usability alone (Barendregt et al., 2006). Based on general observations, the children appeared to be highly engaged with the activities, particularly the quizzes, flashcards, and puzzles, and seemed able to interact with BrainyMinis for extended periods without losing interest. The children also appeared to adapt quickly to the interface after a brief explanation, navigating through the activities with minimal assistance. The vibrant and visually appealing design of the interface seemed to capture and maintain their attention, suggesting a generally positive user experience.

Across modules (quizzes, flashcards, puzzles), respondents displayed indicators of positive emotional engagement, such as smiles, laughter, pointing, verbal excitement, leaning closer to the screen, and staying on task longer than prompted. In child app contexts, such behaviours are often treated as meaningful signs of enjoyment and involvement and can be used alongside performance measures to evaluate whether the experience is emotionally "working" for children (Crescenzi-Lanna, 2020; Barendregt et al., 2006). These observations are significant, as evidence indicates that children's enjoyment is a critical factor influencing the acceptance and continued use of educational applications (Amaefule et al., 2023).

This suggests that the UAT emotional engagement cues provide early acceptance evidence that BrainyMinis' interaction style can sustain attention and willingness to continue, which is critical for repeated exposure and learning opportunities in young children.

The positive engagement obtained from this UAT is consistent with past studies that reported the effect of supportive interface elements, such as warm and appealing visuals, friendly characters, and playful feedback, can increase positive emotions and motivation, which may support deeper engagement in learning tasks (Um et al., 2012; Plass et al., 2014; Heidig et al., 2015).

Despite the positive engagement, the UAT of this study also revealed moments of emotional friction, visible through frowning, pausing, repeated miss clicks, confused faces, pushing the laptop away, asking the teacher for help, as well as losing interest. This pattern aligns with past studies that reported children's emotional responses can shift quickly when interaction becomes ambiguous or when feedback is unclear, and that these affective signals are important indicators of where design refinement is needed (Crescenzi-Lanna, 2020; Barendregt et al., 2006).

Besides emotional friction, the UAT also discovered another challenge, in particular, language barriers, as a few children struggled to fully understand English instructions, highlighting the need for language localization to enhance accessibility. The reward system was well-received, with children showing increased motivation to complete tasks in anticipation of earning rewards. Additionally, feedback from the supervising teacher was positive, acknowledging Brainyminis as an engaging and educational tool for reinforcing classroom learning. Nonetheless, the teacher recommended incorporating introductory learning content before each activity and adding short video tutorials to guide the children, ensuring they fully understand the objectives and mechanics of the games before beginning.

7. CONCLUSION

The BrainyMinis web application was developed to address the limited availability of interactive learning resources in early childhood education, particularly in mathematics. Traditional methods, such as lecture-style instruction and static worksheets, may not consistently engage young learners or accommodate their individual learning paces, potentially limiting their understanding of foundational mathematical concepts. Observations from formative testing suggest that BrainyMinis may support engagement and exploration through interactive activities, such as flashcards, quizzes, and puzzles, and that the Learning Module allows

children to interact with mathematical concepts in a self-paced, play-based context. The Performance Tracking Module appeared to provide parents and teachers with tools to monitor learning milestones and support individualized learning. These findings indicate that BrainyMinis has potential as a tool for promoting interactive and adaptable learning experiences for preschool children.

Looking ahead, further development could extend the educational scope of BrainyMinis to include subjects beyond mathematics, such as language, science, and the arts, in order to provide a more holistic learning experience. Incorporating video tutorials may also support children in navigating activities and understanding task requirements, particularly for those who may need additional guidance. Additionally, implementing bilingual capabilities could improve accessibility for children with varying language backgrounds. Future work will involve systematic evaluation of these enhancements, including empirical studies to assess their effectiveness in supporting learning outcomes and engagement.

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